Title:

Self Powered Intelligent Pedestrian Protection System Using Dynamic Safety Barrier and Energy Harvesting Zebra Crossing

PROJECT REPORT

By

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ABSTRACT

The Self-Powered Intelligent Pedestrian Protection System Using Dynamic Safety Barrier and Energy Harvesting Zebra Crossing is designed to enhance pedestrian safety at zebra crossings by addressing the increasing risks caused by speeding vehicles and negligent driving behavior. Conventional systems rely only on traffic signals, road markings, and driver compliance, which are often insufficient to prevent accidents.

The proposed system introduces an intelligent and autonomous solution that integrates multiple sensors to detect pedestrian presence, traffic signal status, vehicle proximity, and emergency vehicle sirens. Based on real-time inputs, a motorized dynamic safety barrier is automatically controlled to restrict or allow vehicle movement at the crossing, thereby providing physical protection to pedestrians.

A key feature of the system is the incorporation of piezoelectric energy harvesting technology embedded within the zebra crossing. Mechanical energy generated from pedestrian footsteps and vehicle movement is converted into electrical energy, which is conditioned, stored, and reused to power the system components. This enables self-powered and energy-efficient operation without dependence on external power sources.

The system also includes visual and audible alert mechanisms, an LCD-based countdown display, and a camera module for monitoring and violation detection. Additionally, an emergency vehicle override mechanism ensures that ambulances, police, and fire vehicles can pass without delay.

Overall, the proposed invention presents a decentralized, standalone, and sustainable smart traffic solution that enhances pedestrian safety, improves traffic management, and promotes eco-friendly infrastructure.

OBJECTIVE

1. **To design and develop a self-powered intelligent pedestrian protection system** for zebra crossings that enhances pedestrian safety by preventing accidents caused by speeding or negligent vehicles.

2. **To implement an automated dynamic safety barrier mechanism** that physically restricts vehicle entry into the pedestrian crossing zone when pedestrians are detected.

3. **To integrate multiple sensing technologies** including infrared sensors, ultrasonic sensors, light sensors (LDR), sound sensors, and limit switches for detecting pedestrian presence, vehicle proximity, traffic signal status, and emergency vehicle sirens.

4. **To utilize piezoelectric energy harvesting technology** to generate electrical energy from pedestrian footsteps and vehicle movement on the zebra crossing, enabling self-sustained system operation.

5. **To provide real-time alerts and monitoring mechanisms** using buzzers, LCD displays, and a camera module to warn drivers and record traffic violations.

6. **To incorporate an emergency vehicle override mechanism** that detects sirens from ambulances, police vehicles, or fire trucks and temporarily lowers the barrier to allow safe and uninterrupted passage.

7. **To develop a decentralized standalone traffic safety** unit that operates without reliance on cloud servers, internet connectivity, or centralized traffic control infrastructure.

CHAPTER 1

INTRODUCTION

i) **Field of Invention:**

Embedded Systems, Intelligent Transportation Systems, Smart Traffic Management Systems.

ii) **Focus of Invention:**

The invention aims at ensuring improved safety for pedestrians at zebra crossings. This is achieved through the development of an intelligent system capable of detecting pedestrians and operating a physical safety barrier to prevent vehicles from entering the crossing area during the time of pedestrian movement.

iii) **Core Technology of Invention:**

The invention is based on microcontroller automation using Arduino. The invention also employs several sensors, including ultrasonic sensors, infrared sensors, light sensors using LDR, sound sensors, and limit switches. The invention also employs piezoelectric-based energy harvesting technology, actuator-based barriers, buzzers, and LCD displays.

iv) **Integration:**

The invention is based on the integration of several features, including pedestrian detection, vehicle proximity detection, traffic signal detection, and emergency vehicle detection. The invention is also based on energy harvesting using mechanical pressure on the zebra crossing, which is based on piezoelectric plates. This ensures that the invention is self-powered and does not require cloud connectivity.

The Gap

It exists because drivers exceed speed limits and drive recklessly which creates dangerous conditions for pedestrians at zebra crossings. Drivers ignore traffic signals and fail to decrease their speed at pedestrian crossings which creates situations that lead to accidents, injuries, and fatalities. Common zebra crossings depend on traffic signals and road markings and signboards to show drivers the correct way to proceed while lacking any system that would protect pedestrians from harm.

Multiple current systems work to improve pedestrian security through their use of sensor technology for traffic signal system management and their implementation of alert systems

which use flashing lights and alarms. The systems depend on driver adherence to rules because they lack a system which would actively stop vehicles from driving into pedestrian crossing zones during crossing times.

Current traffic safety systems need ongoing external power sources together with centralized traffic monitoring systems for their full functionality. The systems operate in areas which lack dependable power supply or network access while their installation costs and operational expenses rise.

The Novelty

The Self-Powered Intelligent Pedestrian Protection System works to protect pedestrians through its active system which protects people crossing zebra crossings. The invention uses various sensing methods to track current traffic conditions. The sensors identify traffic signal status, pedestrian presence, approaching vehicle distance, and emergency vehicle siren sounds. The system utilizes these inputs to operate a motorized dynamic safety barrier which raises or lowers to control vehicle access at the crossing area.

The system introduces its main innovation through energy harvesting technology which it incorporates into zebra crossings. The crossing surface contains piezoelectric sensors which generate electrical energy from pedestrian footsteps and vehicle movement. The system operates as a self-powered energy-efficient infrastructure unit because it processes energy through its conditioning system before storing and reusing it to power its components.

The system includes visual and audible warning systems which use buzzers and an LCD countdown display to show pedestrians their remaining time to cross. Emergency vehicle detection systems enable ambulances, police, and fire vehicles to drive through by temporarily lowering the barrier. The system includes a camera module which records video footage of vehicles that breach crossing regulations.

The proposed invention establishes a pedestrian safety system which operates independently through decentralized technology while maintaining complete energy efficiency and intelligent functionality. The system uses sensing technology to detect emergency situations while it enforces safety rules through automated enforcement mechanisms and it generates renewable energy to create safer zebra crossings.

CHAPTER 2
LITERATURE SURVEY

A. Patent Literature

S.No	Title	Publication Type	Summary	Gap Identified	Differentiating Methodology
1	Smart Pedestrian Crossing System	Patent	Describes automated pedestrian crossings where traffic lights are controlled using sensors to allow pedestrians to cross safely.	The system relies only on traffic signals and driver compliance, with no physical barrier for pedestrian protection	The proposed invention introduces a dynamic safety barrier that physically stops vehicles during pedestrian crossing, improving safety.
2	Piezoelectric Road Energy Harvesting System	Patent	Describes embedding piezoelectric materials in roads to generate electricity from vehicle movement and road vibrations.	The generated energy is typically used for general lighting or grid supply and is not integrated with pedestrian safety infrastructure.	The proposed system directly integrates piezoelectric energy harvesting with the pedestrian protection system, making the zebra crossing self-powered.
3	Intelligent Traffic Monitoring and Violation Detection System	Patent	Uses cameras and monitoring systems to detect traffic violations and record vehicles that break traffic rules.	These systems require external power and centralized monitoring, increasing deployment cost..	The proposed invention is self-powered and decentralized, operating using harvested energy and embedded control.

B. Non - Patent Literature

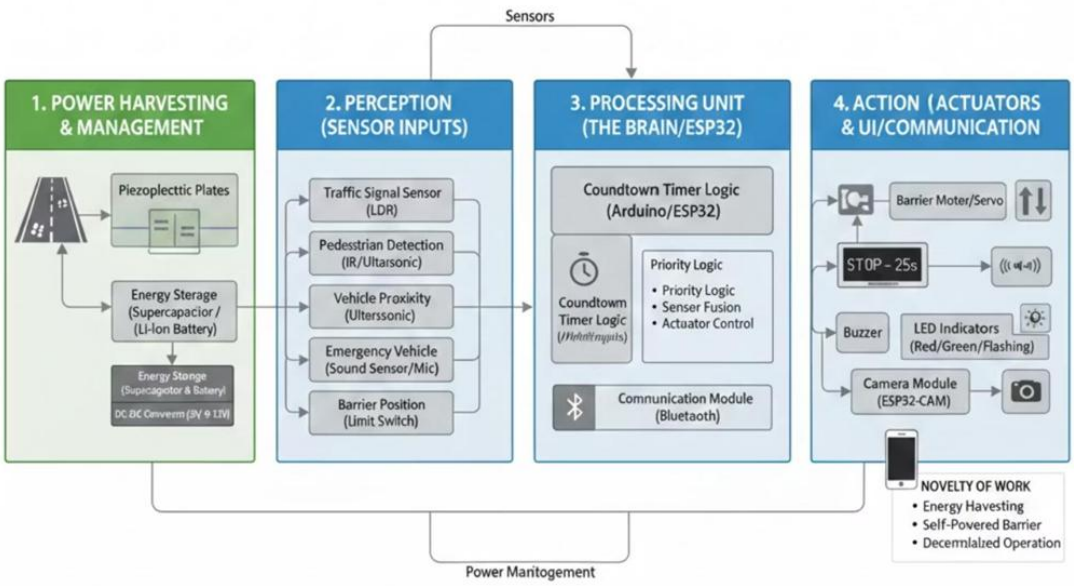
S.No	Title	Publication Type	Summary	Gap Identified	Differentiating Methodology
1	Smart Pedestrian Crossing Using Sensor-Based Traffic Control	Research Paper	Uses IR sensors or cameras to detect pedestrians and adjust traffic lights accordingly to allow safe crossing.	The system relies only on traffic signal changes and does not actively stop vehicles from entering the crossing area.	The proposed system introduces a motorized safety barrier along with multi-sensor detection to physically enforce pedestrian protection.
2	Piezoelectric Energy Harvesting for Smart Road Applications	Journal Article	Studies the generation of electrical energy from mechanical stress produced by vehicles on piezoelectric materials embedded in roads.	The research focuses mainly on energy generation and storage but does not apply the energy to real-time pedestrian safety systems.	The invention uses harvested energy to power the barrier system, sensors, warning lights, and monitoring components.
3	IoT-Based Smart Traffic Monitoring System	Conference Paper	Implements traffic monitoring using cameras, sensors, and internet connectivity for data analysis and traffic control.	Requires internet connectivity and cloud infrastructure, making it unsuitable for decentralized or remote locations.	The proposed invention functions as a standalone intelligent system without cloud or internet dependency.

CHAPTER 3

SYSTEM OVERVIEW

The Self-Powered Intelligent Pedestrian Protection System functions as an independent safety system which safeguards zebra crossing areas. The system utilizes its integrated sensors to track four elements which include traffic signal status and pedestrian presence and approaching vehicles and emergency vehicle sirens. The microcontroller uses sensor data to operate a motorized safety barrier which controls vehicle access while protecting pedestrians at the crossing. The zebra crossing system uses embedded piezoelectric plates to collect mechanical energy from pedestrian footsteps and vehicle movement which the system converts and stores to power its components. The system operates as a self-sufficient pedestrian protection system through its integrated sensors and actuators and energy harvesting modules.

Through this coordinated operation, the system provides a self-powered, intelligent, and autonomous pedestrian protection mechanism that enhances road safety while maintaining efficient traffic flow.



Stepwise Working Principle:

1. **Traffic Signal Detection**
   A Light Dependent Resistor (LDR) sensor monitors the state of the traffic signal. When the signal changes to yellow or indicates pedestrian crossing time, the system prepares to activate the safety mechanism.

2. **Barrier Deployment and Position Verification**
   The microcontroller activates the motorized barrier through a motor driver. The barrier rises to physically block vehicle entry into the zebra crossing zone.

3. **Vehicle Proximity Monitoring**
   Ultrasonic sensors continuously monitor the distance of approaching vehicles near the crossing. If a vehicle moves too close to the barrier or attempts to violate the crossing zone, the system activates safety alerts.

4. **Alert and Violation Detection Mechanism**
   When a violation is detected:
   A buzzer generates an audible alarm.
   A camera module captures an image of the approaching vehicle for monitoring and enforcement purposes.

5. **Pedestrian Crossing Information Display**
   An LCD display provides real-time information to both pedestrians and drivers by showing messages such as “STOP – PEDESTRIANS CROSSING” along with a countdown timer indicating the remaining time available for safe crossing.

6. **Emergency Vehicle Override**
   A sound sensor detects the siren signals of emergency vehicles such as ambulances, police vehicles, and fire trucks. Upon detection, the system temporarily lowers the barrier to allow the emergency vehicle to pass safely without delay.

7. **Energy Harvesting and Storage**
   Piezoelectric plates embedded within the zebra crossing convert mechanical pressure from pedestrian footsteps and vehicle movement into electrical energy. The generated AC voltage is converted to DC using a bridge rectifier and filtered through capacitors. The conditioned energy is then stored in a supercapacitor or rechargeable battery.

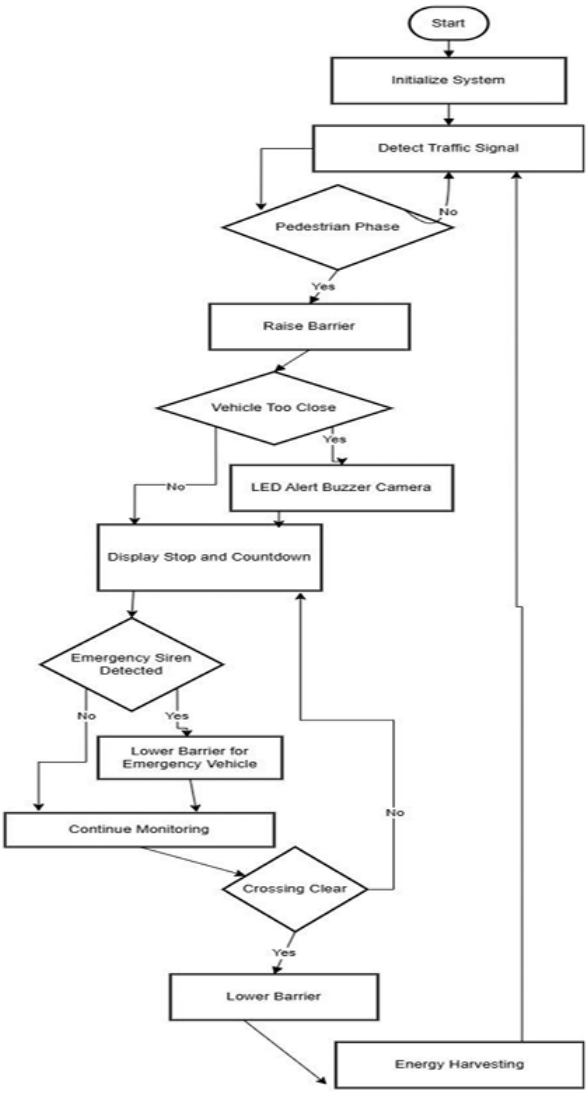
8. **Power Management and System Control**

   A DC–DC converter regulates the stored electrical energy to provide stable voltage levels required for the microcontroller, sensors, display units, and motor operation. The microcontroller processes sensor inputs and synchronizes barrier control, alert systems, and energy usage.

9. **Post-Crossing Safety Check**

   Even after the traffic signal turns green, pedestrian detection sensors verify that no pedestrians remain on the crossing. If pedestrians are still present, the barrier remains active until the crossing area is clear.

Through this coordinated operation, the system provides a self-powered, intelligent, and autonomous pedestrian protection mechanism that enhances road safety while maintaining efficient traffic flow.



CHAPTER 4
HARDWARE DESIGN

The system integrates hardware components, sensors, actuators, energy harvesting modules, and control logic to detect pedestrian and vehicle activity, manage traffic barriers, and provide real-time alerts.

Hardware Components:

• Microcontroller:

Arduino/ESP32 for processing sensor inputs and controlling actuators.

• Sensors:

1. **IR sensor** → Detects the presence of a person near the crossing area. Sends signal to close the barrier and ensure pedestrian safety.
2. **Ultrasonic sensor** → Measures distance of approaching vehicles using sound waves. Triggers alerts or barrier control when a vehicle is too close.
3. **Sound sensor** → Detects high-frequency sounds like ambulance sirens. Gives priority to emergency vehicles by opening the barrier immediately.
4. **Limit switch** → Detects whether the barrier is fully open or fully closed. Prevents over rotation of the motor and ensures accurate positioning.
5. **Piezoelectric sensors** → Generate electrical energy from mechanical pressure applied on the zebra crossing.

• Actuators:

1. **Servo/DC motor** → Controls the physical movement of the barrier (open/close). Operates based on signals from the microcontroller.
2. **Relay / Motor driver** → Acts as a switch to control high-power devices like motors. Protects the microcontroller by handling higher voltage/current loads.

• Output Devices:

1. **LCD display** → Displays system messages. Provides real-time feedback to users at the crossing.
2. **Buzzer** → Produces sound alerts for warnings (e.g., vehicle too close). Enhances safety by giving audible notifications.

• **Energy Harvesting & Storage:**

1. **Piezoelectric plates** → Generate electrical energy from pressure (mechanical energy) caused by vehicles. Acts as a renewable energy source for the system.
2. **Bridge rectifier & capacitor** → Converts AC output from piezo sensors into usable DC voltage. Capacitor smooths fluctuations and stabilizes the output.
3. **Supercapacitor / Li-ion battery** → Stores harvested energy for later use. Provides backup power when main supply is unavailable.

Software Logic :

• Sensors are read by the microcontroller and barrier control logic for safety is administered.

• Internal timers and software counters control pedestrian crossing countdown, elapsed time for task completion, as well as an alarm timer. There is also an emergency vehicle detection that allows autosaving of barrier for barrier override.

• Energy management software allows for stable operation using stored harvested energy.

CHAPTER 5
IMPLEMENTATION AND RESULTS

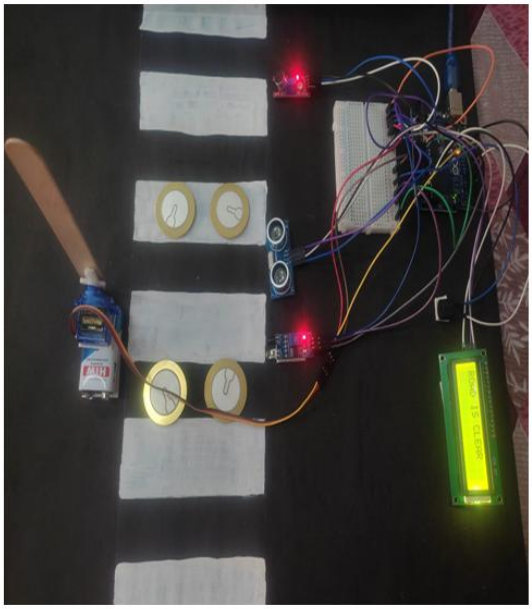
The system was built as a prototype and experiments conducted to evaluate its real-time operation by simulating some traffic scenarios. Through this it was decided whether the various sensors and the alert mechanisms shown were very good or need to be improved. It was also checked whether the energy harvesting unit was being successful and demonstrating the efficiency.

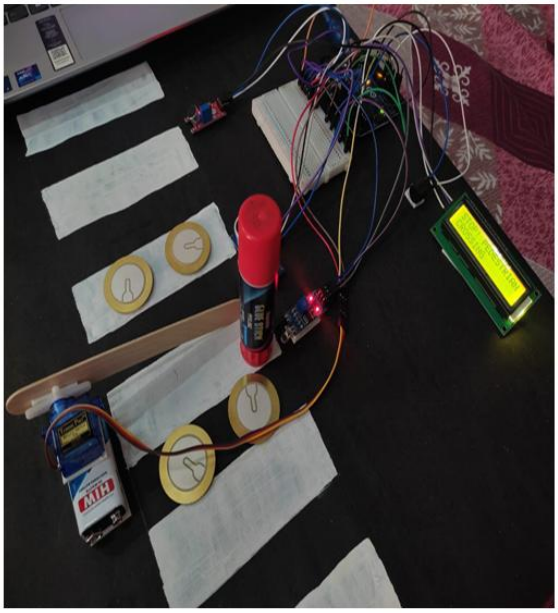
Prototype Testing

A small prototype had been built with the microcontroller, the different sensors, a barrier which responded to the motor and the output devices. Various test scenarios were conducted to validate system performance.

Test Case 1: Pedestrian Detection

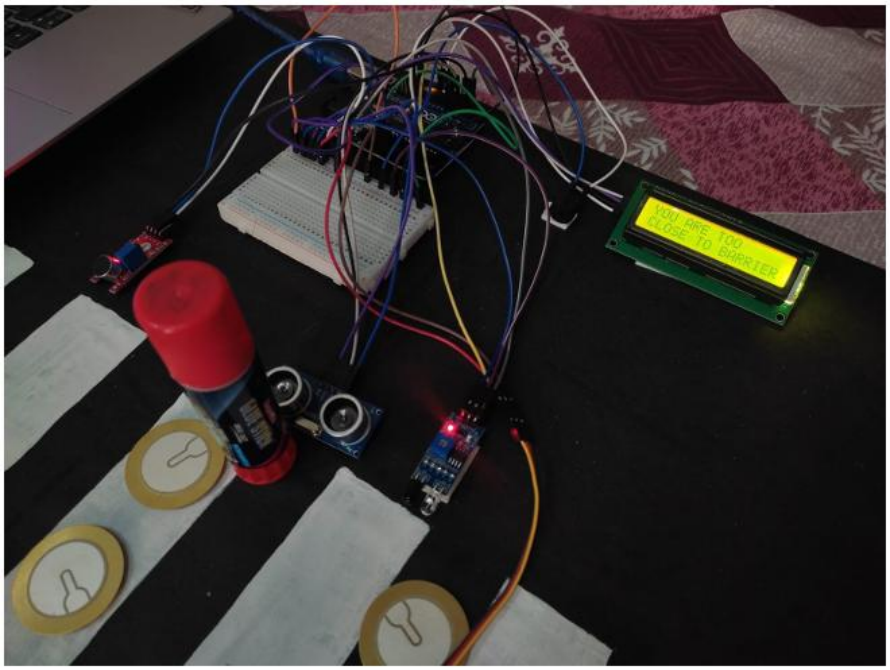
The IR / ultrasonic sensors were used to check if the sensors can detect when a person is coming close to the crossing area. All the sensors were working and on detection of a person the barrier was lowered and the time captured by the timer was given as the countdown so that the pedestrian knows when to cross the road safely.





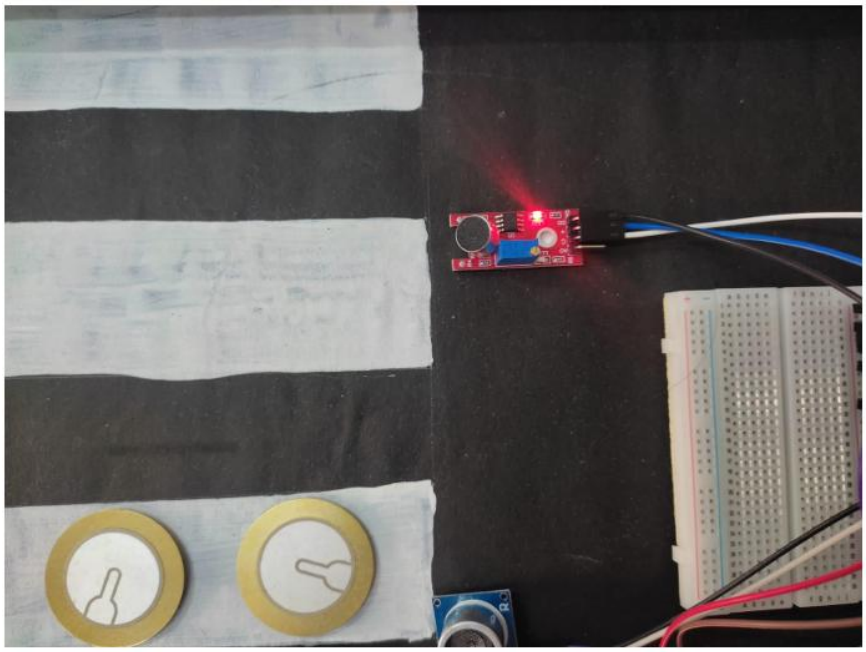
Test Case 2: Vehicle Proximity Detection

The ultrasonic sensor was checking the covered the distance of approaching vehicles. When a vehicle was detected to be getting too close to the crossing area during a pedestrian phase, all the output devices were working to give loud horns and display warnings. The image of the vehicle was taken using the camera module which was used as an output to the system.



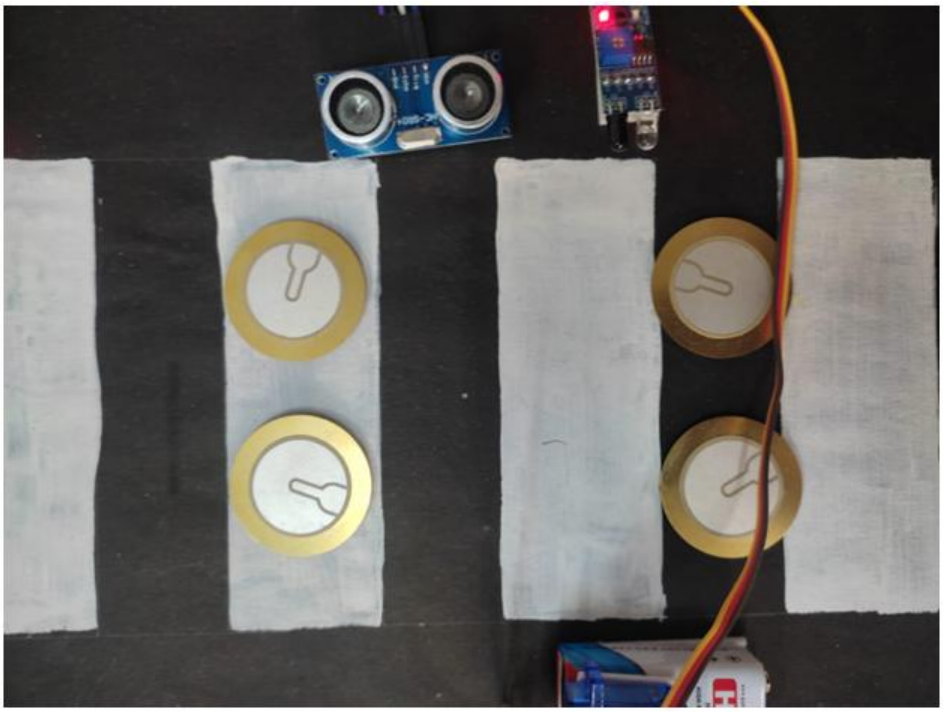
Test Case 3: Emergency Vehicle Detection

The sound sensors detected the emergency vehicle honk and when the system was tested, the barrier was lowered and restriction of the system was set to OFF so that the vehicle could cross easily.



Test Case 4: Energy Harvesting Validation

When a person pushed pressure on the piezoelectric plates they produced small electricity. The electricity produced was done to the rectifier and was stored in the capacitor / battery system to check whether the system self-powered or not.



Results and Observations

The experimental tests were able to validate that:

• The sensors were able to detect the traffic signals, pedestrians, and approaching vehicles.

• The automated barrier system was able to respond correctly to the control signals.

• The visual and audio alert systems were able to alert the vehicles and pedestrians.

• The emergency vehicle detection was able to trigger the override.

• The piezoelectric energy harvesting module was able to generate electricity.

SAMPLE CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>
// I2C address 0x27 is standard for JHD 162A
LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo gateServo;
// Pin Definitions
const int irSensor = 2;
const int soundSensor = 4;
const int trigPin = 6;
const int echoPin = 7;
const int buzzer = 8;
const int servoPin = 9;
// Timer for Siren
unsigned long sirenStartTime = 0;
const long sirenHoldTime = 5000;
void setup() {
pinMode(irSensor, INPUT);
pinMode(soundSensor, INPUT);
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
pinMode(buzzer, OUTPUT);
gateServo.attach(servoPin);
lcd.init();
lcd.backlight();
gateServo.write(0); // Start Open
}
void loop() {
// 1. Get Ultrasonic Distance
long duration;
int distance;
digitalWrite(trigPin, LOW);
```

```
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance = duration * 0.034 / 2;
// 2. Read Sensors
int personDetected = digitalRead(irSensor);
int sirenDetected = digitalRead(soundSensor);
// Update siren timer if sound is heard
if (sirenDetected == HIGH) {
sirenStartTime = millis();
}
lcd.clear();
// 3. LOGIC HIERARCHY
// PRIORITY 1: AMBULANCE (Master Override)
if (millis()- sirenStartTime < sirenHoldTime) {
gateServo.write(0); // OPEN for ambulance
lcd.setCursor(0, 0);
lcd.print("AMBULANCE ON THE");
lcd.setCursor(0, 1);
lcd.print("WAY! MAKE WAY");
}
// PRIORITY 2: PEDESTRIAN DETECTION (Person is present)
else if (personDetected == LOW) {
gateServo.write(90); // CLOSE for cars
lcd.setCursor(0, 0);
lcd.print("STOP! PEDESTRIAN");
lcd.setCursor(0, 1);
lcd.print("CROSSING");
}
// PRIORITY 3: VEHICLE PROXIMITY ALERT
else if (distance > 0 && distance < 10) {
gateServo.write(0);
```

```
lcd.setCursor(0, 0);
lcd.print("YOU ARE TOO");
lcd.setCursor(0, 1);
lcd.print("CLOSE TO BARRIER");
}
// DEFAULT: ROADCLEAR
else {
gateServo.write(0); // OPEN for cars
lcd.setCursor(0, 0);
lcd.print("ROAD IS CLEAR");
}
// 4. BUZZER CONTROL (Independent safety beep)
if (distance > 0 && distance < 10) {
digitalWrite(buzzer, HIGH);
} else {
digitalWrite(buzzer, LOW);
}
delay(200);
}
```

CONCLUSION

The proposed Self-Powered Intelligent Pedestrian Protection System Using Dynamic Safety Barrier and Energy Harvesting Zebra Crossing successfully addresses the critical issue of pedestrian safety at zebra crossings. The system integrates multiple sensing technologies, automated barrier control, and energy harvesting mechanisms to create a smart, efficient, and reliable safety solution. Through the design and implementation of the prototype, it has been demonstrated that the system is capable of accurately detecting pedestrian presence, monitoring vehicle proximity, identifying traffic signal conditions, and responding to emergency vehicle sirens. The motorized dynamic safety barrier effectively restricts vehicle movement during pedestrian crossing phases, thereby providing physical protection rather than relying solely on driver compliance.A significant contribution of the proposed system is the integration of piezoelectric energy harvesting technology, which enables the conversion of mechanical energy from pedestrian footsteps and vehicle movement into usable electrical energy. This allows the system to operate as a self-powered and energy-efficient unit, reducing dependency on external power sources and making it suitable for deployment in areas with limited infrastructure.

The inclusion of real-time alert mechanisms such as LED indicators, buzzers, and LCD displays enhances awareness among both pedestrians and drivers. Additionally, the incorporation of a camera module for monitoring and violation detection strengthens the enforcement aspect of the system. The emergency vehicle override feature ensures that critical services such as ambulances, police, and fire vehicles can pass without delay, maintaining a balance between safety and traffic efficiency.

Overall, the system provides a decentralized, standalone, and intelligent solution for pedestrian safety, combining automation, sensing, and sustainable energy utilization. The experimental validation results confirm that the proposed system operates effectively under various simulated conditions and has the potential to significantly reduce accidents at pedestrian crossings.

Future Scope -

The system can be further enhanced by integrating advanced technologies such as:

- AI-based image processing for automatic license plate recognition and violation tracking
- IoT-based connectivity for remote monitoring and data analytics
- Solar energy integration along with piezoelectric harvesting for improved power efficiency
- Deployment in real-world urban environments with scalability for smart city applications
- Integration with centralized traffic management systems for coordinated traffic control

PROJECT VIDEO LINK

Link of project demonstration Video:

https://drive.google.com/file/d/1PLnyqrELPTGwBE3i7XoFrh_Uo4A9X68C/view?usp=sharing